

VZ200 ROM Font

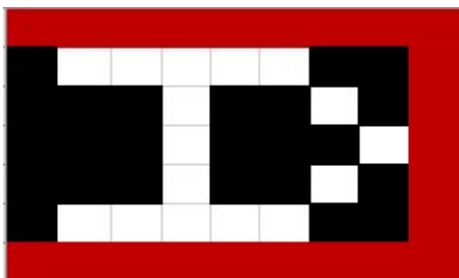
Tips and Tricks

The VZ Rom (version 2.0) contains 320 bytes of font data starting at address 3B94h (15252 decimal). This allows for character output to a plotter, and can also be used to draw pixels to the screen.

Each character takes 5 bytes, arranged in an 8x5 grid. A snippet of the ROM disassembly is below:

3B94	C1 BE A2 AE B1	DEFB	0C1H,0BEH,0A2H,0AEH,0B1H	;@
3B99	83 ED EE ED 83	DEFB	83H,0EDH,0EEH,0EDH,83H	!A
3B9E	80 B6 B6 B6 C1	DEFB	80H,0B6H,0B6H,0B6H,0C1H	;B
3BA3	C1 BE BE BE DD	DEFB	0C1H,0BEH,0BEH,0BEH,0DDH	;C

On inspecting the bit pattern, the characters are rotated 90 degrees. Instead of appearing upright, each character appears to be laying on its side. For example the "A" character is defined like this:



The red border around the character is just used here to show the 8x5 grid.

It is worth noting that only 64 ASCII characters are defined – letters A to Z, numbers 0 to 9, and some symbols. So any logic to draw text and symbols to the screen in graphics mode using the ASCII code will need to adjust for the characters available.

The program below demonstrates a way to convert a string entered by the user into graphics. It adjusts for the space character (line 55) but not much else. Eg it doesn't handle numbers as I wanted to keep it simple

BASIC PROGRAM

```
10 CLS:INPUT A$
20 MODE(1)
30 COLOR 3
40 FOR L=1 TO LEN(A$)
50   C=ASC(MID$(A$,L,1))-65
55   IF C < 0 GOTO 140
60   FOR R=0 TO 4
70     M=1
80     FOR BT=0 TO 7
90       B = PEEK(15257+R+C*5)
100      IF (B AND M) <> M THEN
SET(R+L*7,BT)
110      M = M*2
120    NEXT BT
130  NEXT R
140 NEXT L
150 GOTO 150
```

DISPLAY

